ESE 5410 Midterm Practice Problems - Solutions

Problem 1. Answer the following questions.

1. Explain whether the following scenario is a classification or regression problem. Finally, provide n and p: In order to predict housing prices in Philadelphia, we collect a set of data on 1000 houses. For each house we record number of bedrooms, number of bathrooms, and school scores. We are interested in understanding which factors affect housing prices.

   Regression, $n = 1000$, $p = 3$ (number of bedrooms, number of bathrooms, school scores).

2. Consider a binary classification problem for which you know that the conditional class probabilities are given by:

$$p_1(x) = \Pr(Y_i = 1 | X_i = x) = \frac{x^2 - \frac{1}{2}}{x^2 + \frac{1}{2}} \text{ and } p_0(x) = \Pr(Y_i = 0 | X_i = x) = 1 - p_1(x).$$

   Note that these probabilities are only defined in the range $x \leq -\frac{1}{\sqrt{2}}$ and $x \geq \frac{1}{\sqrt{2}}$. Find the optimal Bayesian classifier.

   The Bayesian classifier is given by the condition: $C(\mathbf{x}) = 1$ when $p_1(\mathbf{x}) \geq 1/2$; 0 otherwise. Hence, from

$$\frac{x^2 - \frac{1}{2}}{x^2 + \frac{1}{2}} \geq \frac{1}{2}$$

   we obtain that $C(\mathbf{x}) = 1$ for x $\geq \frac{\sqrt{6}}{2}$ or $x \leq -\frac{\sqrt{6}}{2}$; 0 otherwise

3. (True or False) In the medical case, the cost of False Negatives is much higher than the cost of False Positives.

   True, it is better to be safe and inform someone they have an illness/disease/injury when they really don't (False Positive) than to let them think they are fine when they actually have an illness/disease/injury (False Negative)

Problem 2. Assume that 'nature' behaves according to the following linear model:

$$Y = \beta_0 + \beta_1 X + \varepsilon,$$

where ε is a Gaussian random variable $\mathcal{N}\left(0, 8s^2\right)$.

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{5} = \{(-1,-3), (0,-2), (1,0), (2,2), (3,3)\}.$$

1. Compute the estimates of $\hat{\beta}_0$ and $\hat{\beta}_1$ for a linear estimator $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$ using the data in $\mathcal{D}$.

   Using the given equations, we obtain

   $$\hat{\beta}_1 = \frac{(-2)(-3) + (-1)(-2) + (0)(0) + (1)(2) + (2)(3)}{4+1+0+1+4} = 1.6$$

   $$\hat{\beta}_0 = -1.6$$

2. Compute the RSS, RSE, and R^2 of the linear fit.

   $RSS = 0.4$, $RSE = \sqrt{\frac{2}{15}}$, $R^2 = \frac{64}{65}$

3. Compute the 95% confidence intervals for β_0 and β_1 as a function of s.

   Using the given equations, we obtain

   $$SE(\hat{\beta}_0) = \sqrt{\frac{12s^2}{5}} = \frac{2\sqrt{15}s}{5}, SE(\hat{\beta}_1) = \sqrt{\frac{8s^2}{10}} = \frac{2\sqrt{5}s}{5}$$

   Hence, the 95% confidence intervals are

   $$\beta_0 \in [-1.6 - 2 \times \frac{2\sqrt{15}s}{5}, -1.6 + 2 \times \frac{2\sqrt{15}s}{5}]$$

   $$\beta_1 \in [1.6 - 2 \times \frac{2\sqrt{5}s}{5}, 1.6 + 2 \times \frac{2\sqrt{5}s}{5}]$$

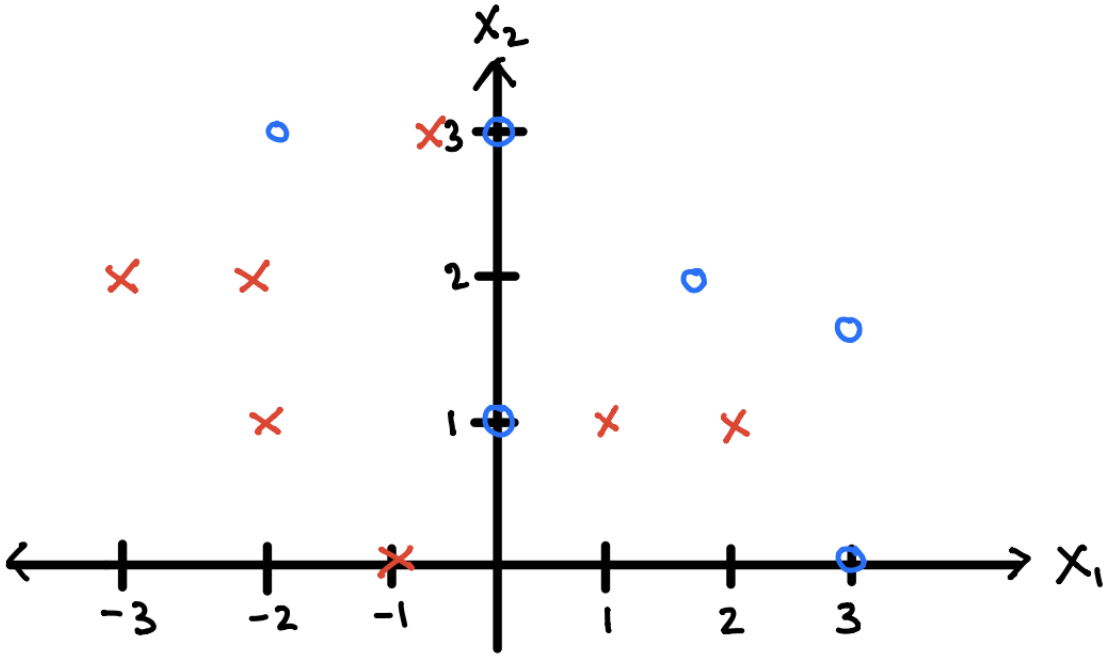
4. For what values of s would you reject the null hypothesis $H_0 : \beta_1 = 4$? Use a 95% confidence level in your answer.

   We need β_1 outside the confidence interval $[1.6 - 2 \times \frac{2\sqrt{5}s}{5}, 1.6 + 2 \times \frac{2\sqrt{5}s}{5}]$. This implies that $s < \frac{3\sqrt{5}}{5}$

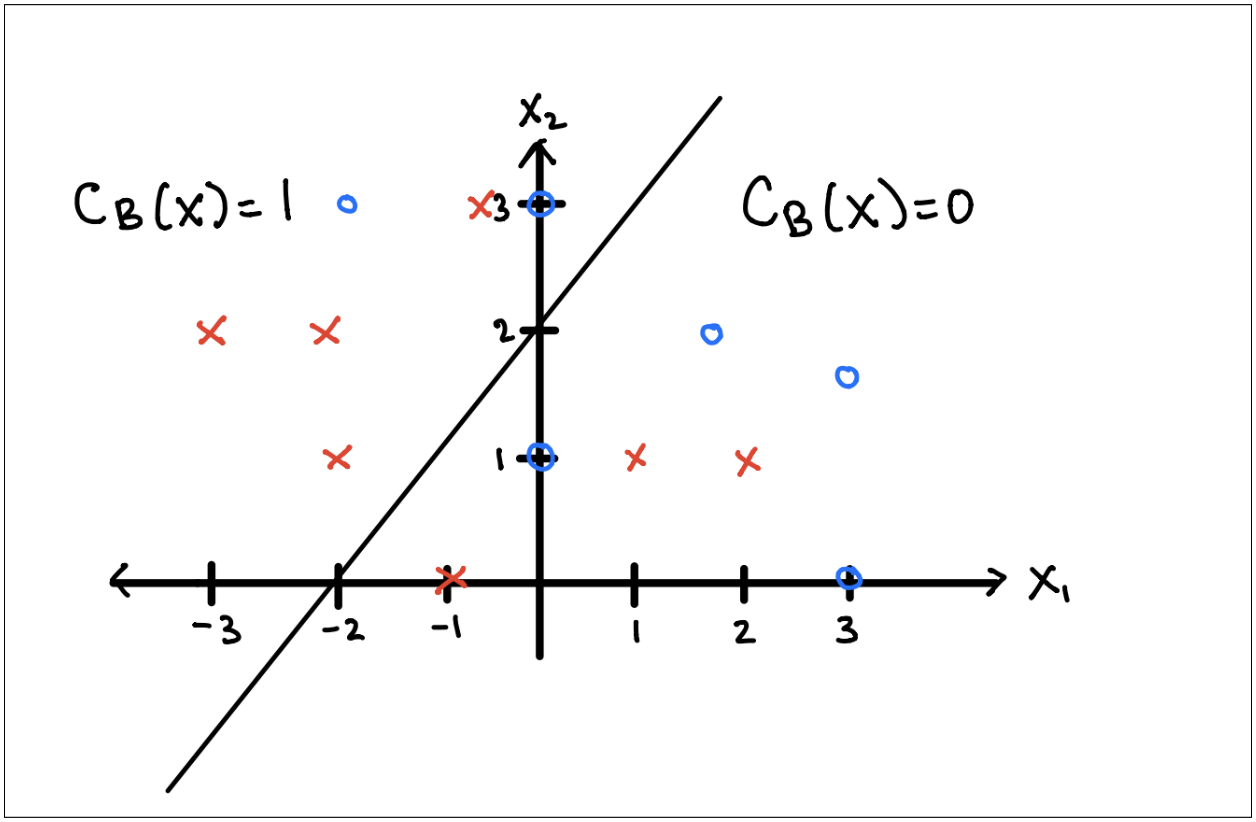
5. For what values of s you *cannot* reject the null hypothesis $H_0 : \beta_0 = 2$? Use a 95% confidence level in your answer.

   We need β_0 to be in the confidence interval $[-1.6 - 2 \times \frac{2\sqrt{15}s}{5}, -1.6 + 2 \times \frac{2\sqrt{15}s}{5}]$. This implies that $s > \frac{3\sqrt{15}}{10}$

Problem 3. Consider the cloud of 13 points in the figure below, where the red points correspond to the label $Y = 1$ and the blue points to the label $Y = 0$. Assume we classify our points using a conditional class probability given by $g_1(x) = \Pr(Y = 0|X = x) = \sigma(2 + x_1 - x_2)$, where σ is the sigmoidal function ($\sigma(z) = e^z/(1 + e^z)$).



1. Draw in the x_1/x_2 plane the decision boundary associated with the Bayesian classifier, denoted by $C_B(\mathbf{x}) \in \{0, 1\}$.



2. Compute the number of the False Negatives, False Positives, and the confusion table of the sigmoidal classifier.

FNs: 3, FPs: 2

Actual Values

Predicted Values	Positive (1)	Negative (0)
Positve (1)	4	2
Negative (0)	3	4

3. Consider another classifier $C_{NB}(\mathbf{x})$ satisfying $C_{NB}(\mathbf{x}) = 1 - C_B(\mathbf{x})$ (i.e., the new classifier flips the labels of the Bayesian classifier). Compute the number of the False Negatives, False Positives, and the confusion table.

FNs: 4, FPs: 4

		Actual Values	
		Positive (1)	Negative (0)
Predicted Values	Positve (1)	3	4
	Negative (0)	4	2

4. Consider a classifier $C_2(\mathbf{x})$ where all outputs are 1, i.e., $C_2(\mathbf{x}) = 1$ for all x. Compute FNR and FPR of this classifier.

$FNR = \frac{\text{False Negative}}{\text{Positive}} = \frac{0}{7} = 0$, $FPR = \frac{\text{False Positive}}{\text{Negative}} = \frac{6}{6} = 1$

Problem 4: Answer the following questions (and make sure you justify your answers):

a. Consider a binary classification task for which you know that the posterior probability is given by

$$p_1(x) = \Pr(Y_i = 1|X_i = x) = \frac{1}{1+x^2} \text{ and } p_0(x) = \Pr(Y_i = 0|X_i = x) = 1 - p_1(x).$$

Find an explicit function for the Bayesian classifier $C(x)$, with the smallest error rate.

$p_1(x) \geq p_0(x)$ on the interval [-1,1], but is less than $p_0(x)$ elsewhere
So, the Bayes optimal classifier is:

$$C(x) = \begin{cases} 1, & \text{if } x \,\epsilon\, [-1,1] \\ 0, & \text{otherwise} \end{cases}$$

b. Explain, in simple words, what the FPR, FNR, and the TPR represent in the context of detecting spam email.

FPR: False Positive Rate for spam detection measures out of all emails that are actually not spam, the amount that are incorrectly classified as spam.
FNR: False Negative Rate for spam detection measures out of all emails that are actually spam, the amount that are incorrectly classified as not spam
TPR: True Positive Rate for spam detection measures out of all emails that are actually spam, the amount that are correctly classified as spam.

c. Describe, in simple words, a situation in which a variable X_2 is strongly correlated to another variable X_1, but X_2 is not directly caused by X_1. In general, can you infer causality from correlation?

Consider the situation where,
X_1 - Hot chocolate sales
X_2 - Gas expense
During winter season, both X_1 and X_2 go up and during summer season, both the variables go down. Due to this relation, it can be said that X_2 is strongly correlated to X_1. However, it is not because of hot chocolate sales that the gas expenses go up. In winters, people usually prefer commuting through their own cars instead of walking or public transportation. Hence, X_2 is not directly caused by X_1. Both X_2 and X_1 are caused by winter season and by one another.

d. Are independent variables always uncorrelated? Are dependent variables always correlated? Justify your answers.

Independent variables infer that X_2 is not dependent on X_1. Since there is no association, they are always uncorrelated.
Dependent variables infer that X_2 is dependent on X_1. Consider the following two cases

$$X_2 = m * X_1$$

$$X_2 = m * X_1^2$$

In both the cases, X_2 is dependent on X_1. In the first case X_2 is also correlated to X_1 - as X_1 increases, X_2 increases if m is positive and as as X_1 increases, X_2 decreases if m is positive.
However in the second case X_2 is not correlated to X_1. X_2 increases with the absolute values of X_1 and not with the actual values of X_1.
Hence, dependent variables may not be always correlated.

e. Consider a variable $Y = X + X^2 + \varepsilon$, where $\varepsilon \sim \mathcal{N}(1,2)$. Provide an expression for the regression function $f(x) = \mathbb{E}[Y|X = x]$. How much is the irreducible error?

ϵ follows a Normal distribution with mean as 1 and variance as 2.

$$f(x) = \mathbb{E}[Y|X = x]$$

$$f(x) = \mathbb{E}\left[x + x^2 + \epsilon\right]$$

$$f(x) = \mathbb{E}[x] + \mathbb{E}\left[x^2\right] + \mathbb{E}[\epsilon]$$

$$f(x) = x + x^2 + 1$$

Therefore, the irreducible error in this case will be the variance of ϵ which from the question is given as 2.

f. Explain, in simple words, how you would use the MSE_{Te} to choose the optimal flexibility for a model.

We need to minimize MSE to obtain the optimal flexibility.

g. *True* or *False*: A very flexible model can lead to overfitting when the Bias term in the Bias-Variance tradeoff is very high. Justify your answer.

False. A very flexible model that leads to overfitting is due to high variance. High bias leads to underfitting.

h. Explain, in simple words, how to build the Bayes optimal classifier using the conditional class probabilities. Is it true that the Bayes optimal classifier always renders the smallest combined number of False Negative and False Positive errors? What about just the smallest number of False Negative errors?

For a given input, the Bayes Optimal Classifier assigns the label that maximizes the conditional class probability. The BOC minimizes the classification error rate overall. Though, this doesn't necessarily mean that the number of False Negatives are minimized. For instance, we could minimize false negatives by always predicting the positive class (no negatives), but this isn't the strategy taken by the BOC.

Problem 5: Consider a *biased* coin with an *unknown* probability of heads, denoted by p.

a. Assume that we perform a coin flip experiment, tossing a coin $n = 3$ times, and obtaining the following sequence of Heads and Tails: H, H, T. What is the probability of observing that particular sequence of heads and tails as a function of the unknown variable p?

$P(H, H, T) = P(H) * P(H) * P(T)$ because coin flips are independent of what has happened previously.
$P(H, H, T) = p * p * (1 - p) = p^2 * (1 - p)$

b. Assume now that we toss a coin 30 times. What is the probability of observing a sequence containing 20 heads and 10 tails (in any order) as a function of the unknown variable p?

Since the order of the sequence does not matter in this case, there are $\binom{30}{20}$ sequences that have 20 heads and 10 tails. $P(\text{20 heads and 10 tails}) = \binom{30}{20} * p^{20} * (1 - p)^{10}$

Problem 6: Answer the following questions:

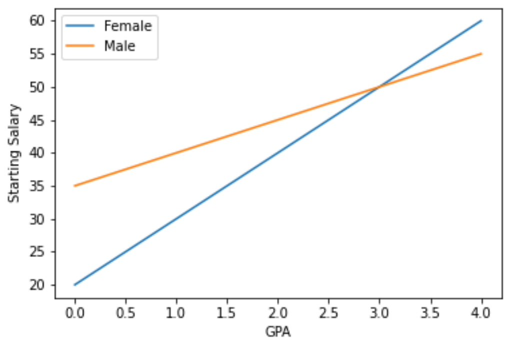
a. Suppose we have a data set with three predictors, X_1=GPA, X_2=Gender (1 for Female and 0 for Male), X_3=Interaction GPA×Gender. The response Y is starting salary after graduation (in thousands of dollars). Suppose we use least squares to fit the model

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \times \text{GPA} + \hat{\beta}_2 \times \text{Gender} + \hat{\beta}_3 (\text{GPA} \times \text{Gender}),$$

and get $\hat{\beta}_0 = 35$, $\hat{\beta}_1 = 5$, $\hat{\beta}_2 = -15$, $\hat{\beta}_3 = 5$. Plot the two regression functions $\hat{Y}(X_1)$ corresponding to the cases X_2=Male and X_2=Female. Plot these two functions for $X_1 \in [0, 4]$.

The two regression functions are given by:

$\hat{Y}_{female} = 20 + 10 * GPA$
$\hat{Y}_{Male} = 35 + 5 * GPA$



b. For what range of values of GPA do Males earn more (on average) than Females?

Males earn more than Females for GPA < 3.0

c. Suppose that, using the same data as in Q2.a., we fit another predictor using the following nonlinear model:

$$\tilde{Y} = \tilde{\beta}_0 + \tilde{\beta}_1 \times \text{GPA} + \tilde{\beta}_2 \times \text{Gender} + \tilde{\beta}_3 (\text{GPA} \times \text{Gender}) + \tilde{\beta}_4 (\text{GPA}^2 \times \text{Gender}).$$

Is the training RSS for this model lower than the training RSS for the previous model.

Yes, since the model is more flexible, the training RSS either stays the same or is reduced

d. If we compute the training RSS for $\hat{Y}$ and $\tilde{Y}$ using ONLY the data points corresponding to Males (discarding those of Females), which RSS is lower?

Because the Gender variable is 0 for males, the training RSS is same for both $\hat{Y}$ and $\tilde{Y}$ if we only use the data points corresponding to Males

e. Would you answers change if we use the test RSS instead of the training RSS?

Yes the answers would change if we use the test RSS instead of training RSS. Test RSS for $\tilde{Y}$ may be higher if it's overfitting the training data. It may also be lower in case $\tilde{Y}$ is a better non-linear approximation to the true model.

Problem 7: Assume that 'nature' behaves according to the following model:

$$Y = \beta_0 + \beta_1 X + \varepsilon,$$

where ε is a Gaussian random variable $\mathcal{N}\left(0, w^2\right)$. Using this model, nature generates a data set

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{5} = \{(-2, -3.5), (-1, -3), (0, 0), (1, 3), (2, 3.5)\}.$$

a. Compute the estimates of $\hat{\beta}_0$ and $\hat{\beta}_1$ for a linear estimator $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$ using the data in $\mathcal{D}$.

It can be seen that $\overline{x} = 0, \overline{y} = 0$

$$\sum x_i^2 = 2 * 1^2 + 2 * 2^2 = 10, \sum x_i y_i = 7 + 3 + 0 + 3 + 7 = 20$$

$$\hat{\beta}_1 = \frac{\sum (x_i - \overline{x})(y_i - \overline{y})}{\sum (x_i - \overline{x})^2} = \frac{\sum x_i y_i}{\sum x_i^2} = \frac{20}{10} = 2$$

$$\hat{\beta}_0 = \overline{y} - \hat{\beta}_1 \overline{x} = 0 - 2 * 0 = 0$$

b. Compute the 95% confidence intervals for β_0 and β_1 as a function of w.

$$SD(\hat{\beta}_1) = \frac{w}{\sqrt{\sum (x_i - \overline{x})^2}} = \frac{w}{\sqrt{10}}$$

$$SD(\hat{\beta}_0) = w\sqrt{\frac{1}{N} + \frac{\overline{x}^2}{\sum (x_i - \overline{x})^2}} = \frac{w}{\sqrt{5}}$$

$$CI : [\hat{\beta}_1 - 2SD(\hat{\beta}_1); \hat{\beta}_1 + 2SD(\hat{\beta}_1)] = [2 - \frac{2w}{\sqrt{10}}; 2 + \frac{2w}{\sqrt{10}}]$$

$$[\hat{\beta}_0 - 2SD(\hat{\beta}_0); \hat{\beta}_0 + 2SD(\hat{\beta}_0)] = [-\frac{2w}{\sqrt{5}}; \frac{2w}{\sqrt{5}}]$$

c. For what values of w can you reject the null hypothesis $H_0 : \beta_1 = 0$?

When β_1 is not in the confidence interval:

$$0 < 2 - \frac{2w}{\sqrt{10}}$$

$$\frac{2w}{\sqrt{10}} < 2$$

$$w < \sqrt{10}$$

d. How would your answers change if the noise ε is a Gaussian $\mathcal{N}\left(1, w^2\right)$?

All the standard deviations would be unchanged, but all the y datapoints will be biased by $+1$. This would result in $\hat{\beta}_0 = 1$ and the associated change in its confidence interval.

Problem 8: Consider a family of classifiers based on flipping a coin with Pr(Head) $= p$, such that we assign a label $\hat{Y} = 1$ if the coin comes out 'Head' and $\hat{Y} = 0$ if the coin comes out 'Tail'.

a. Prove that the ROC of this family of classifiers is a straight line.

An ROC curve is a plot with the false positive rate (FPR) on the x-axis and the true positive rate (TPR) on the y-axis.

$$TPR = \frac{TP}{TP + FN}$$
$$FPR = \frac{FP}{FP + TN}$$

For example, lets say we have x positives and $1 - x$ negatives. We also know that $P(Head) = p$ so,

$$TP = px$$

$$FP = p * (1 - x)$$
$$FN = (1 - p) * x$$
$$TN = (1 - p) * (1 - x)$$

When plugging these equations into TPR and FPR you get

$$TPR = \frac{px}{px + (1 - p)x} = \frac{px}{px + x - px} = \frac{px}{x} = p$$

$$FPR = \frac{p(1 - x)}{p(1 - x) + (1 - p)(1 - x)} = \frac{p}{p + 1 - p} = \frac{p}{1} = p$$

When you plot p against p, you will get a straight line.

b. What is the value of the AUC?

$AUC = \int_0^1 p\,dp = \frac{p^2}{2}$ evaluated between 0 and 1.
$\frac{1^2}{2} - \frac{0^2}{2} = \frac{1}{2}$
AUC for a random classifier is $\frac{1}{2}$ regardless of class proportion.

c. Without any knowledge about the input variable X, can you design a classifier achieving a higher AUC? How about a lower AUC? Justify your answers.

To achieve a higher AUC, we need to increase the TPR and decrease the FPR. To achieve a lower AUC, we must do the opposite which is decreasing the TPR and increasing the FPR.